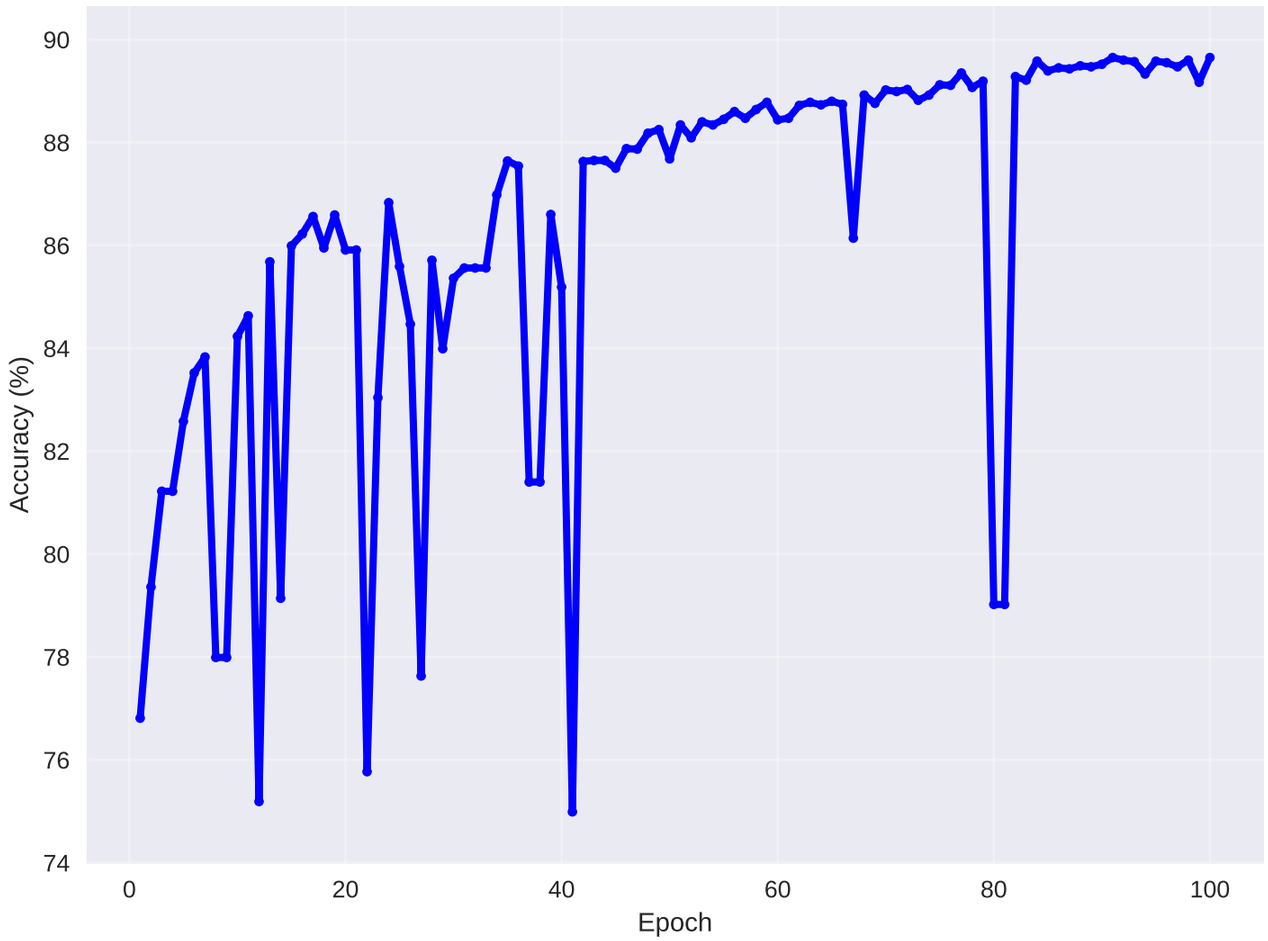
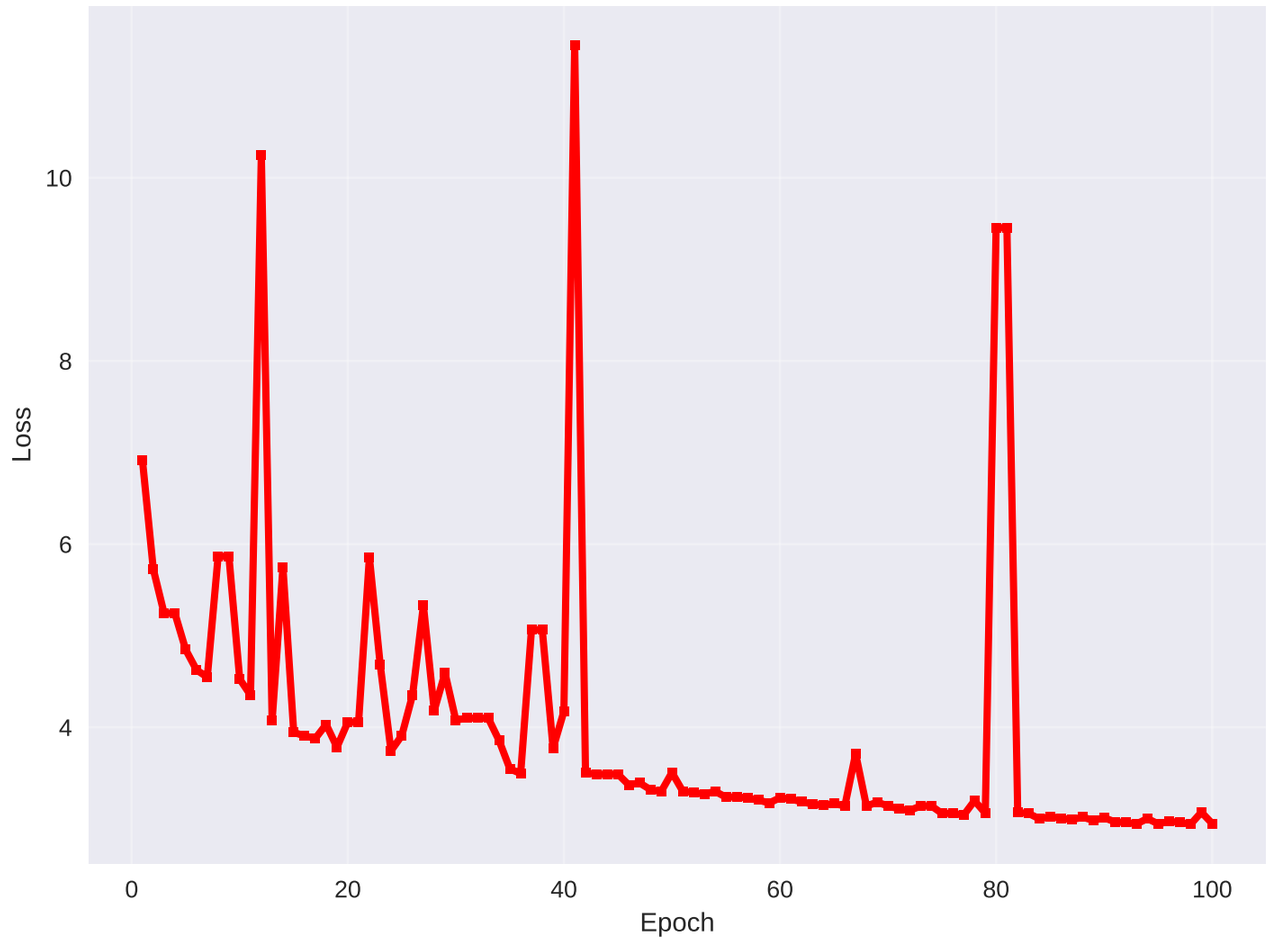
Federated Learning - Main Metrics Overview

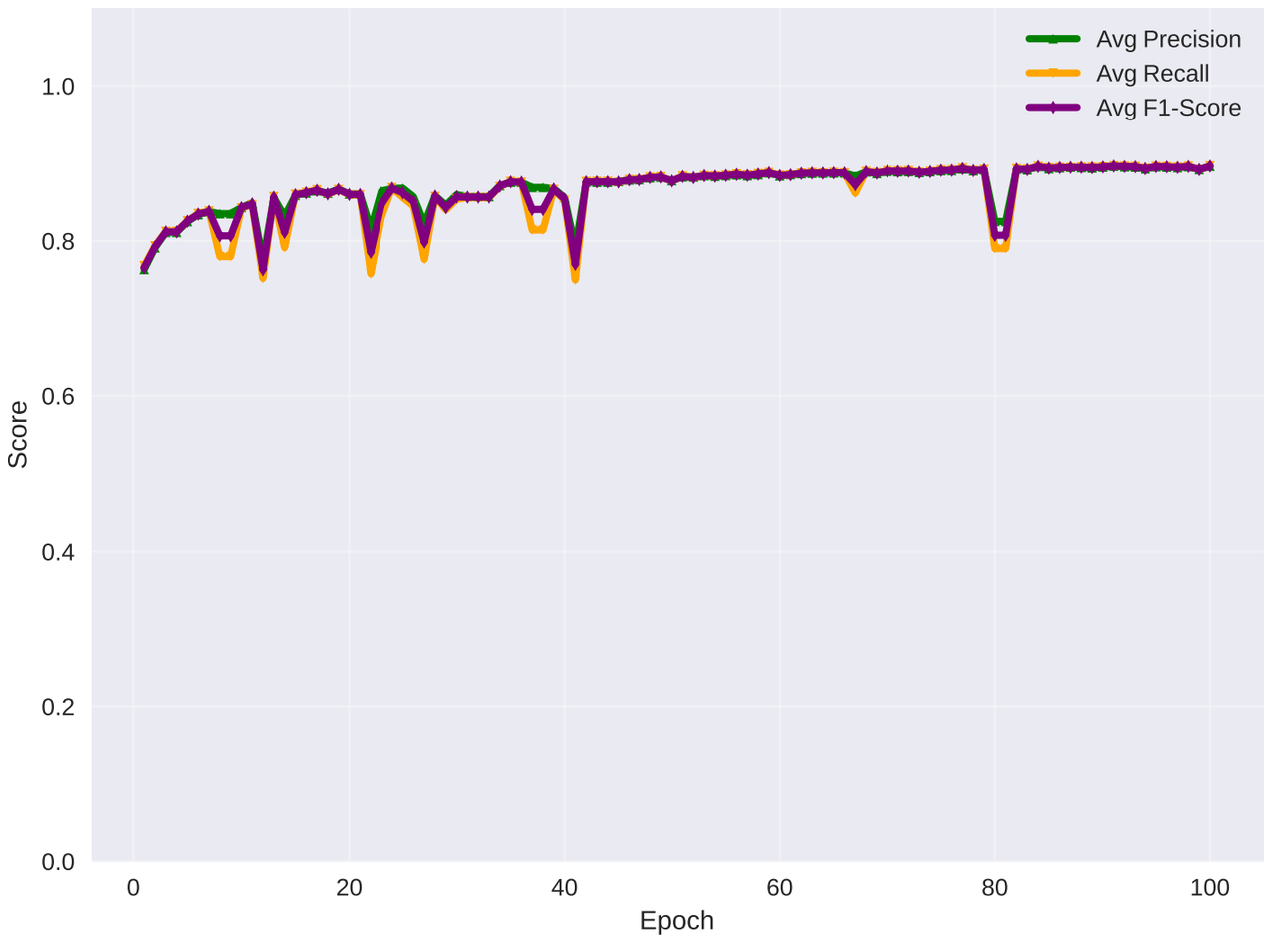
 Global Model Accuracy Evolution



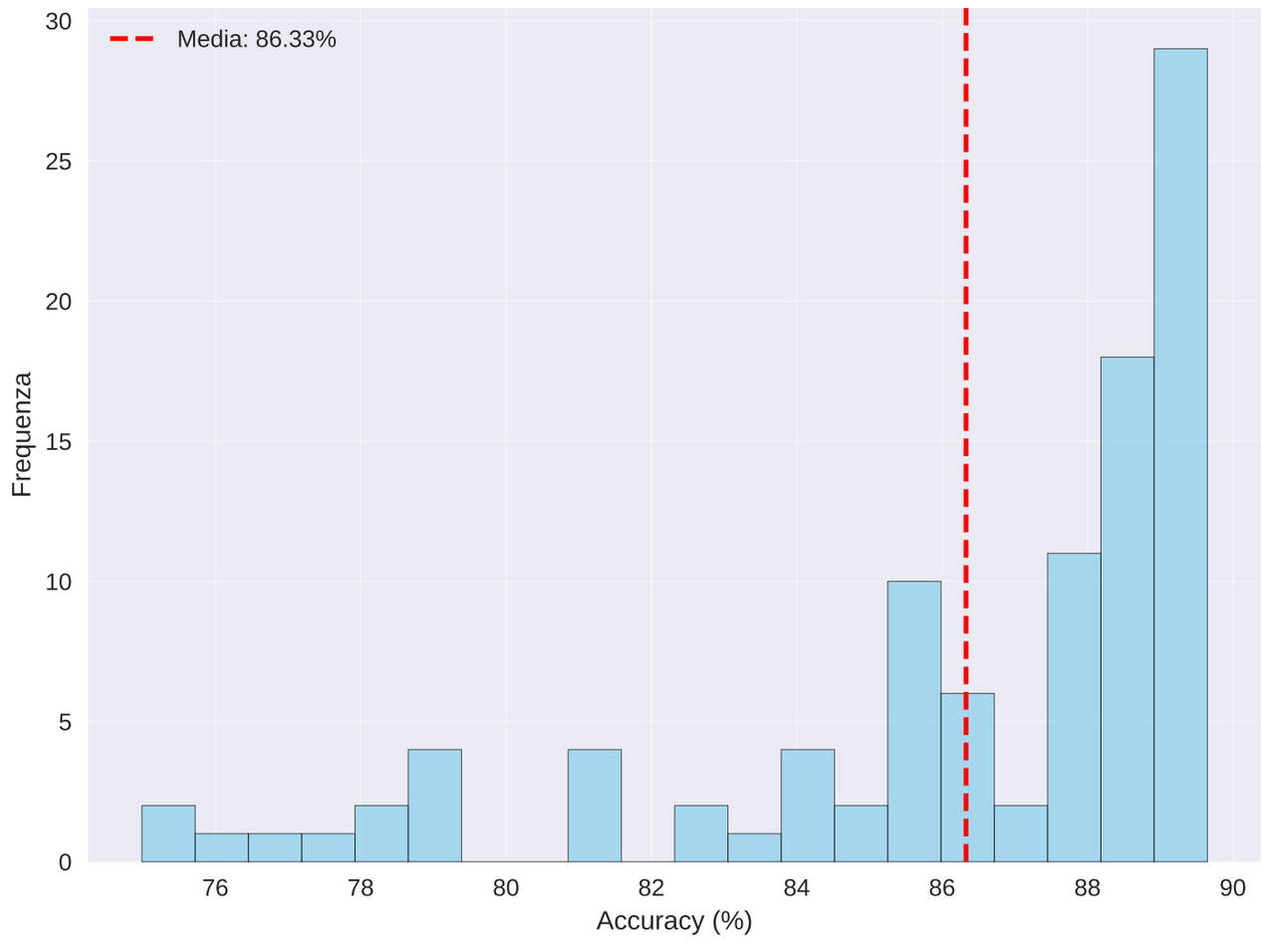
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

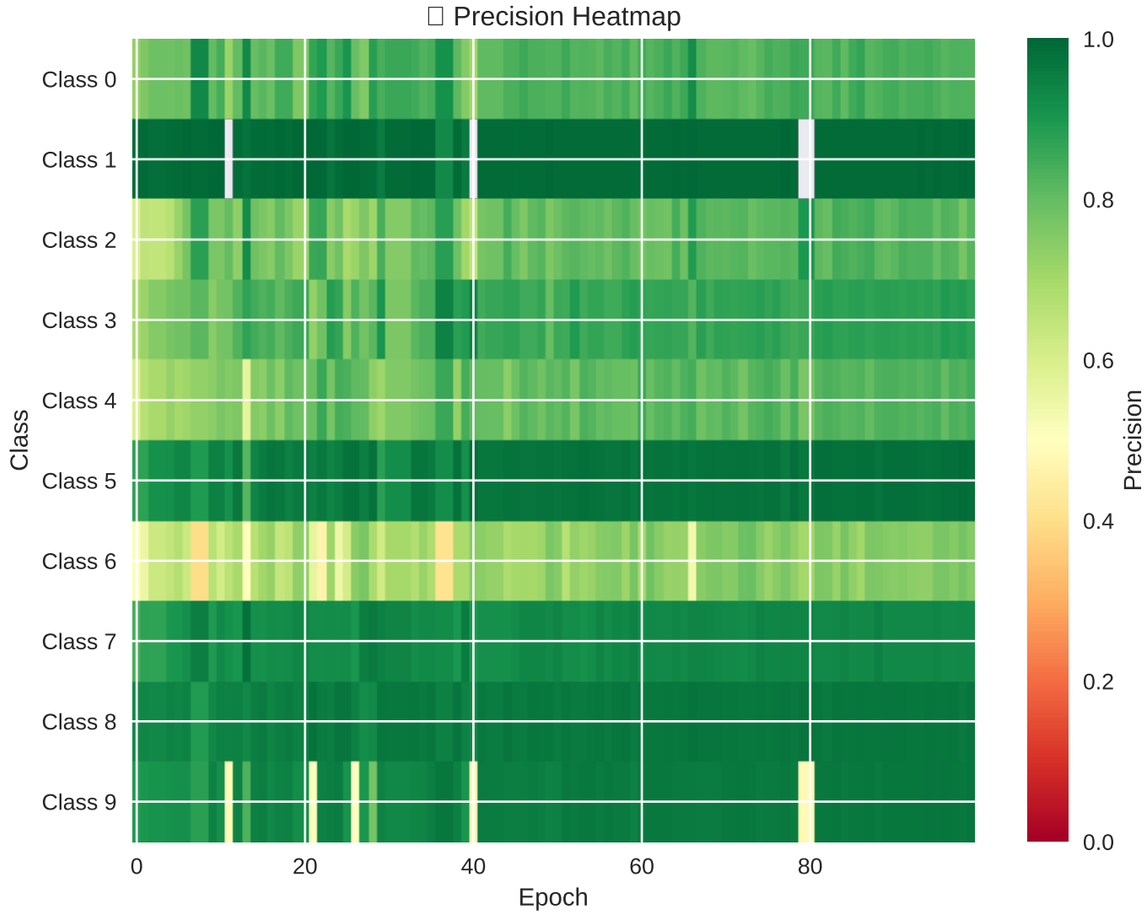


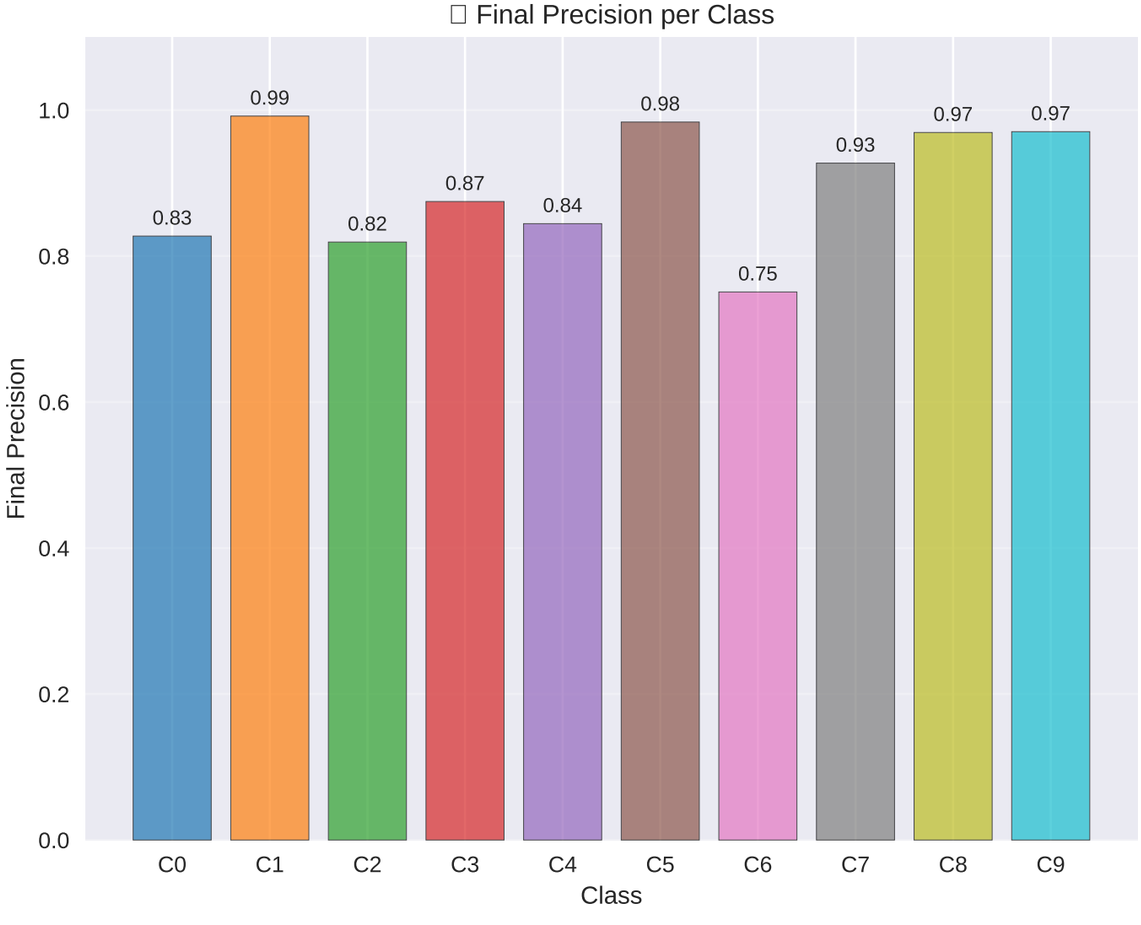
 Distribuzione Accuracy

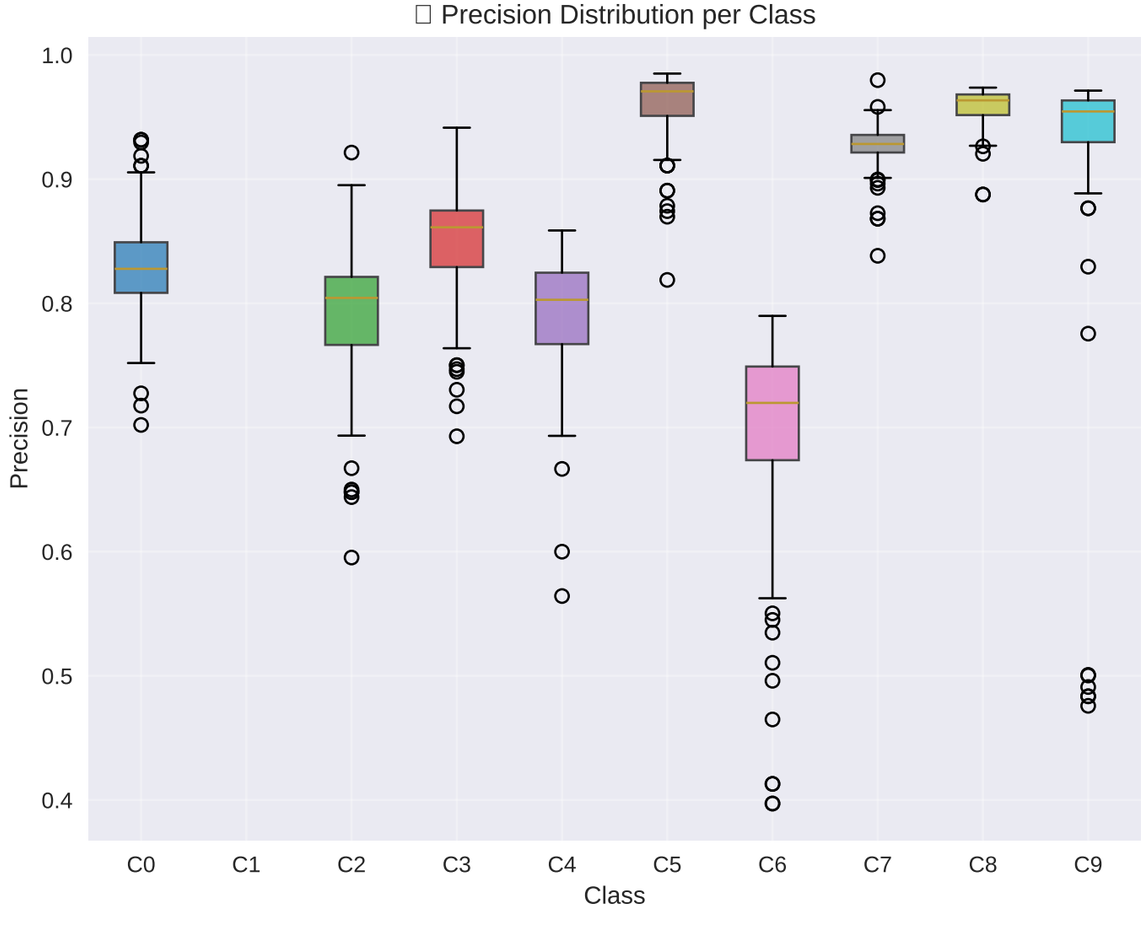


Per-Class Precision Analysis

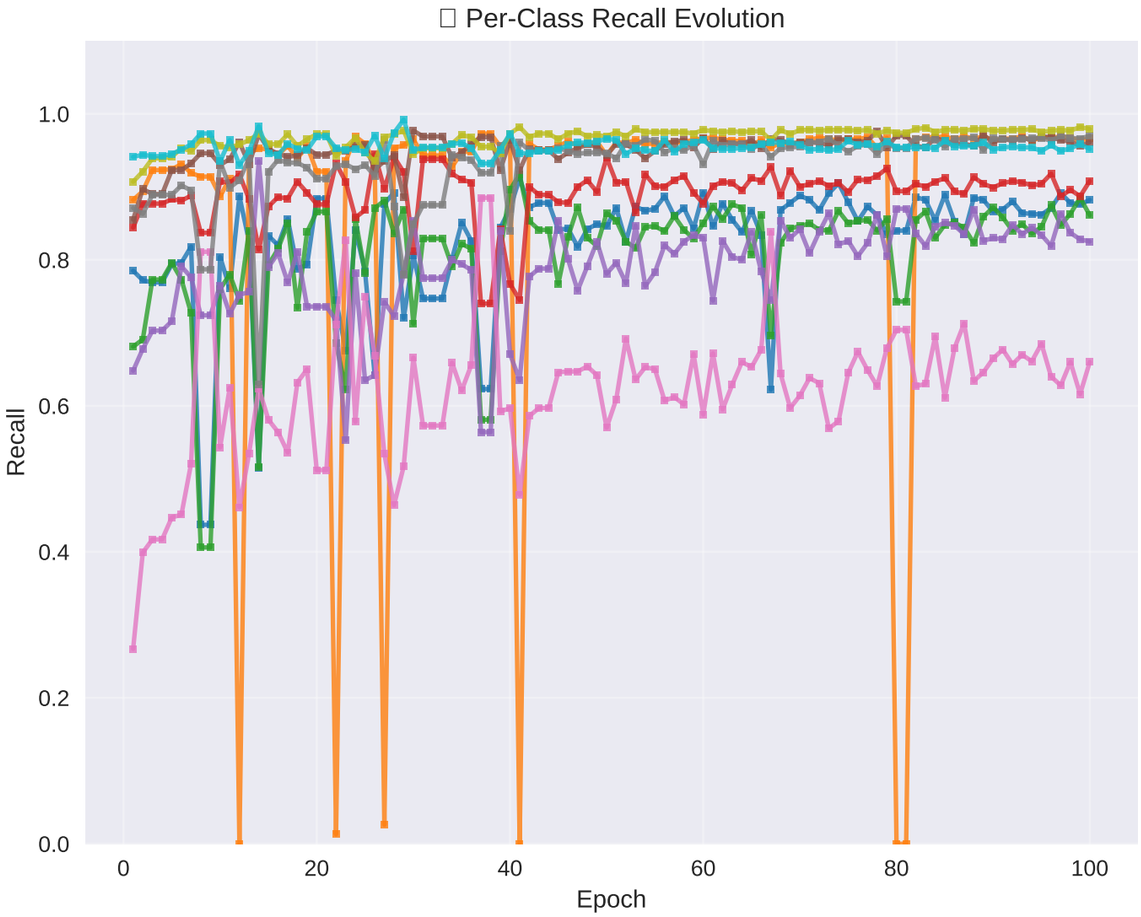


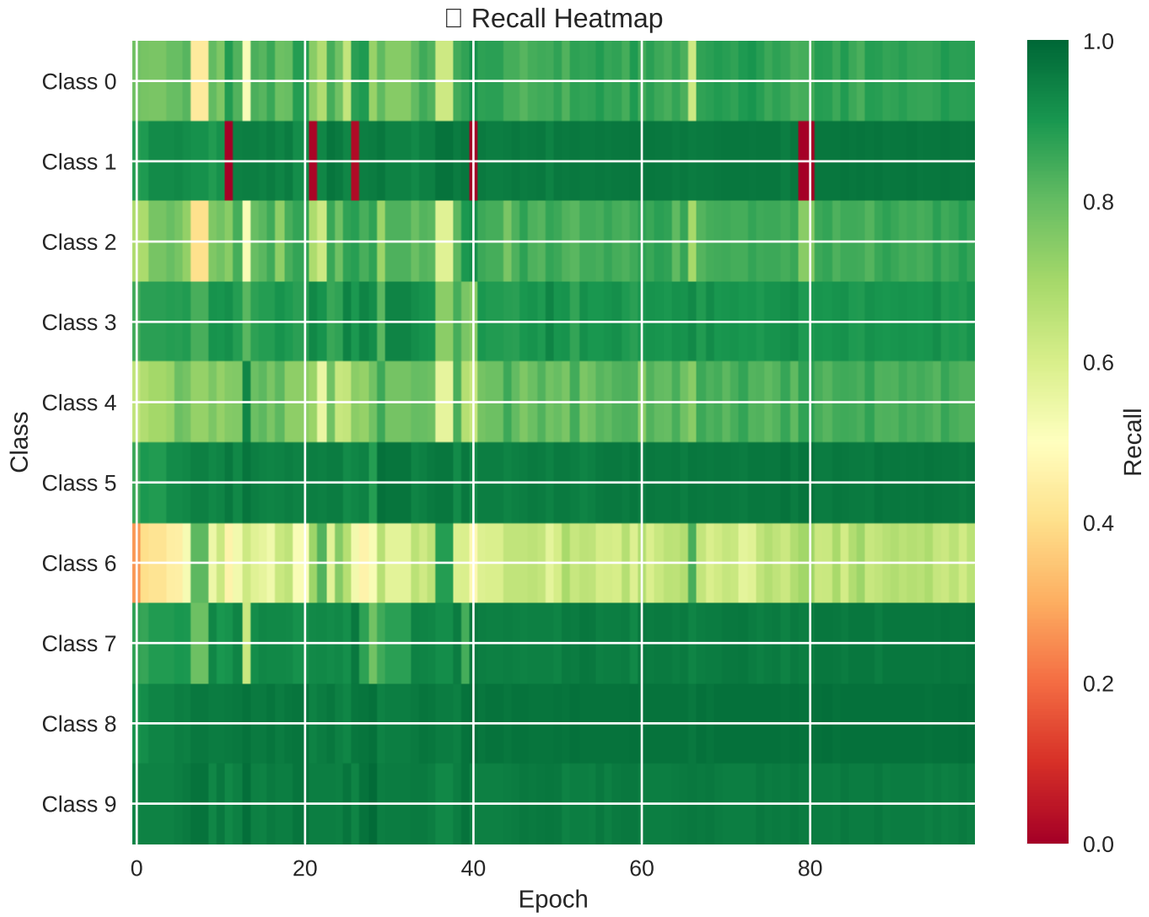


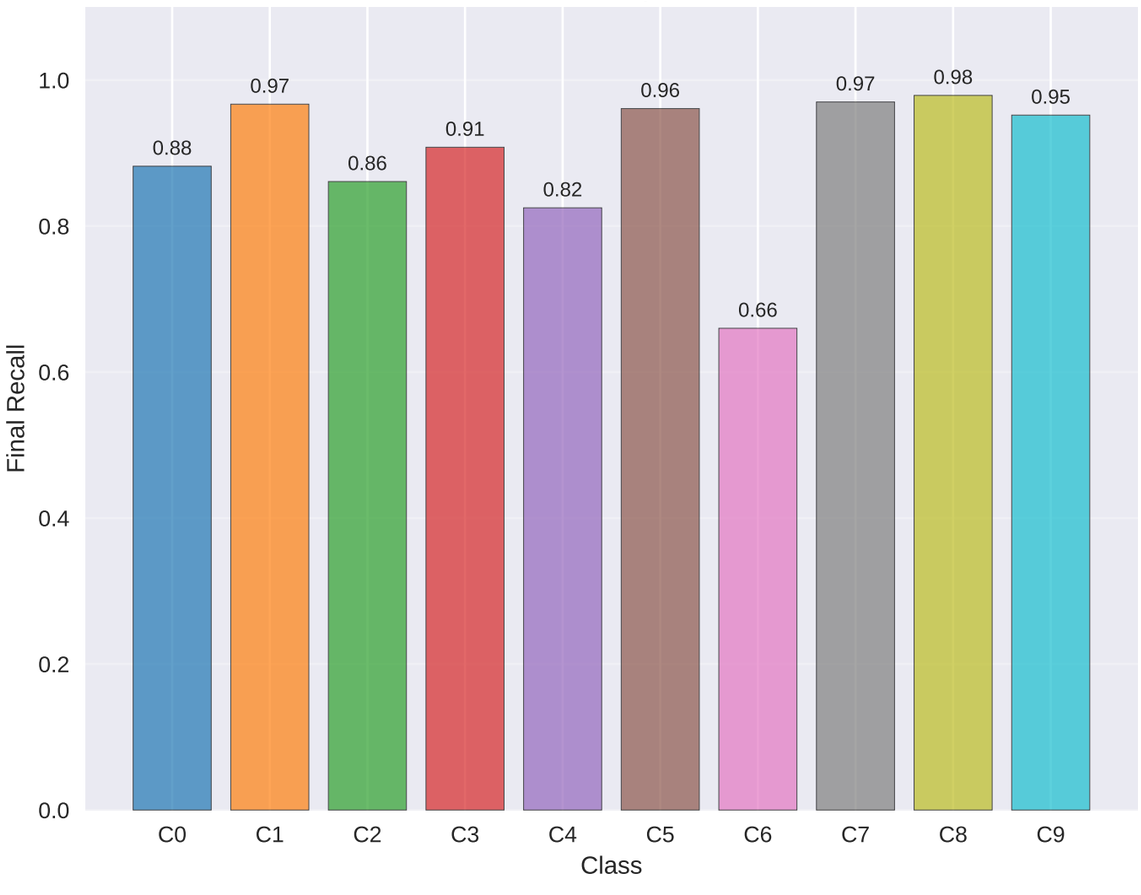


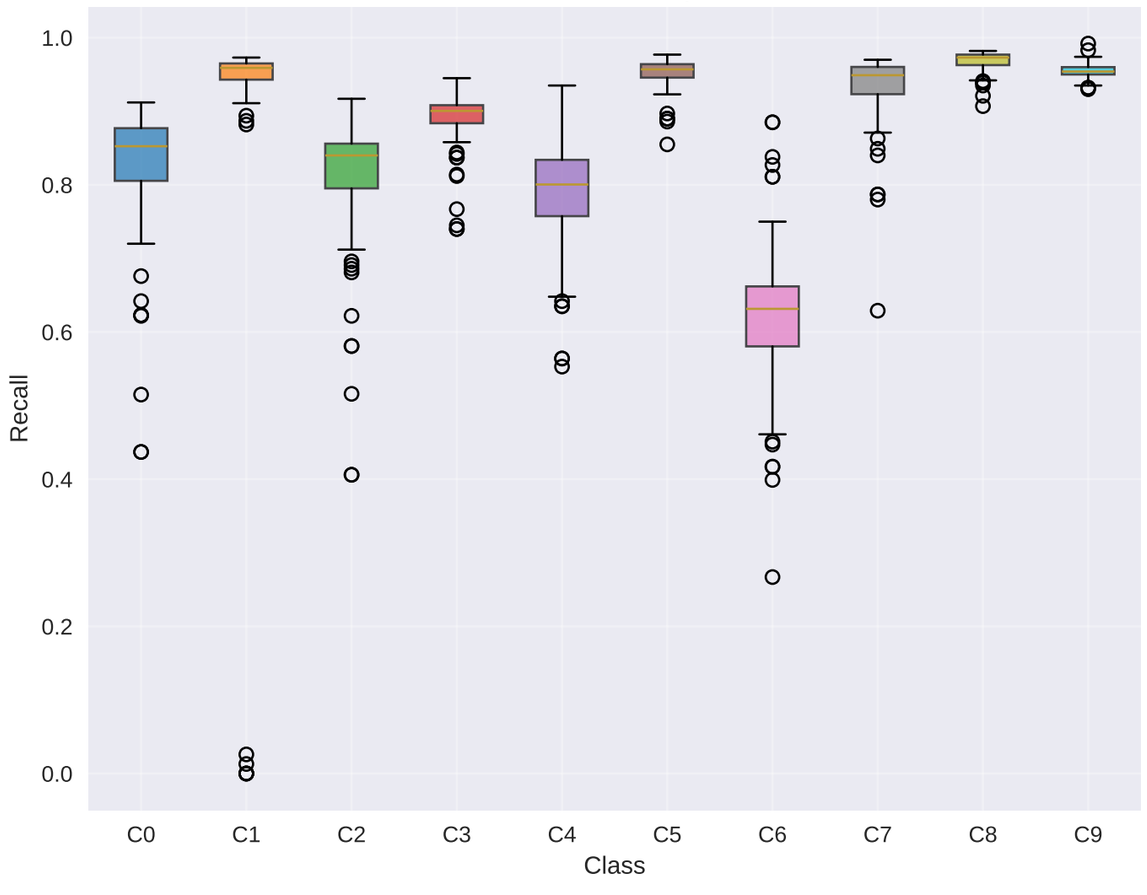


Per-Class Recall Analysis

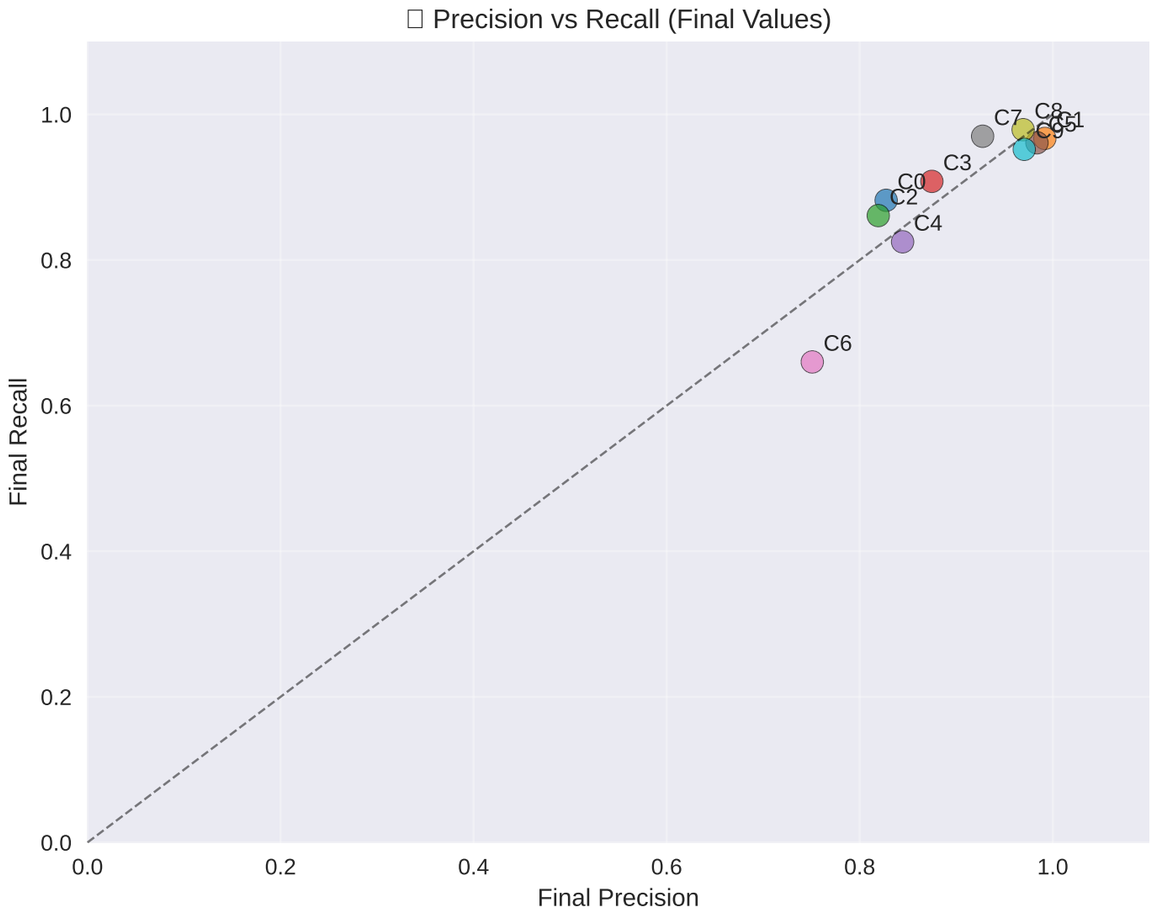






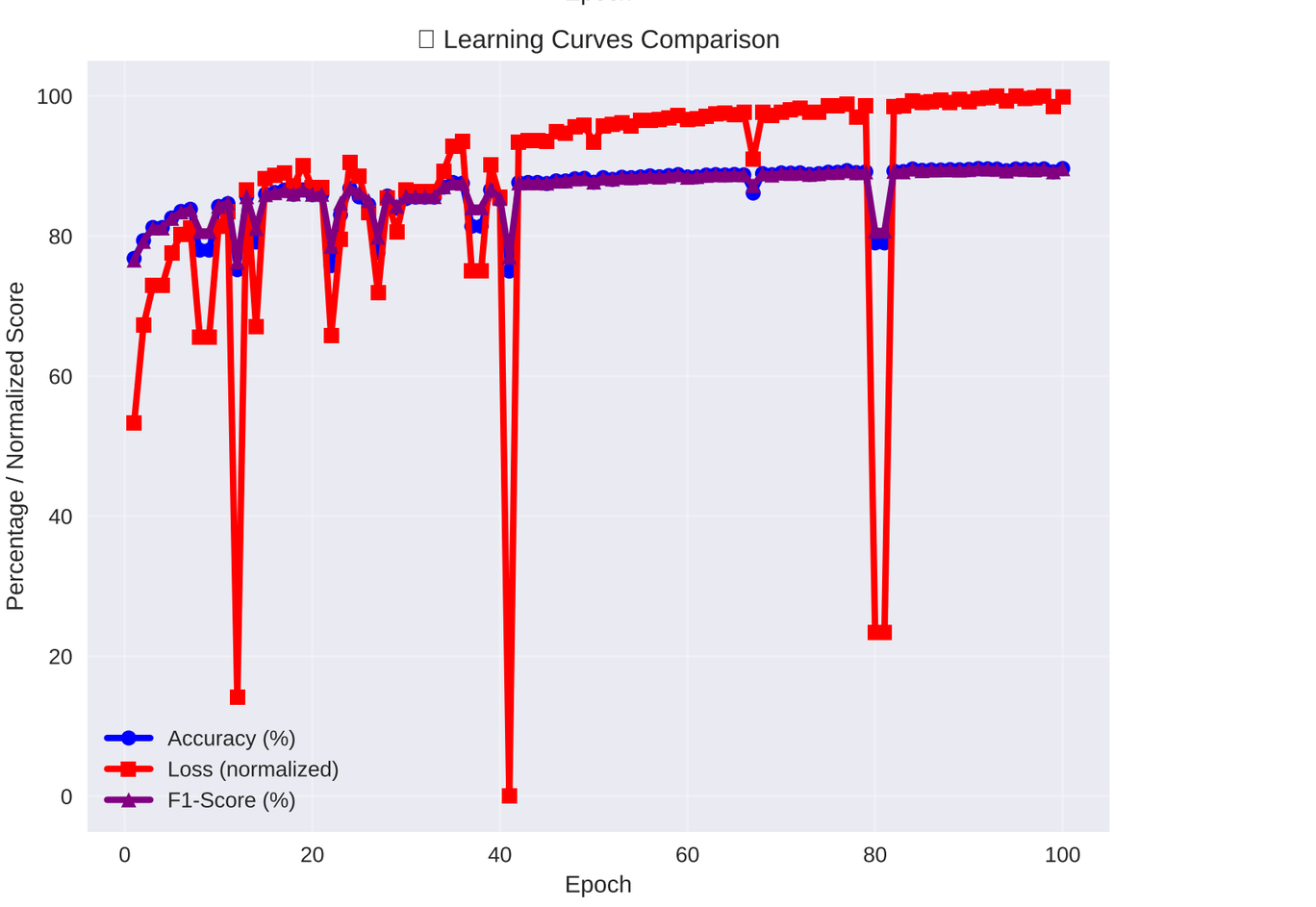


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-10 16:05:25
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 89.65%
• Loss: 2.9483
• Average Precision: 0.896
• Average Recall: 0.896
• Average F1-Score: 0.896

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +12.84%
• Riduzione Loss: +3.9663
• Stabilità Accuracy (CV): 0.0438

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.992)
• Peggiore Precision: Classe 6 (0.751)
• Migliore Recall: Classe 8 (0.979)
• Peggiore Recall: Classe 6 (0.660)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.827, R=0.882, F1=0.854
• Classe 1: P=0.992, R=0.967, F1=0.979
• Classe 2: P=0.819, R=0.861, F1=0.840
• Classe 3: P=0.875, R=0.908, F1=0.891
• Classe 4: P=0.844, R=0.825, F1=0.835
• Classe 5: P=0.984, R=0.961, F1=0.972
• Classe 6: P=0.751, R=0.660, F1=0.703
• Classe 7: P=0.927, R=0.970, F1=0.948
• Classe 8: P=0.969, R=0.979, F1=0.974
• Classe 9: P=0.970, R=0.952, F1=0.961

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
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